

KALASIN ENGINEERING CO., LTD.

PROJECT SPRINT 18K TPA SPRAY DRYER / JET COOKER | REF: x2608003

FACTORY ACCEPTANCE TEST (FAT) – 1-DAY AGENDA REPORT

“Done on FAT (1 Day)” – Focused on Main Control Panel (MCP-01: Chassis C1, C2, C3, C4)

Conforming to IEC 62381 / ISA-RP60.8 Standards | Venue: xDev Engineering Workshop

Document Field	Specification	Document Field	Specification
Document Reference:	KAL-FAT-AGN-2608-C1C4	Issue Date:	08-October-2026
Revision:	Rev. 1.2 (C1-C4 + DI/DO Loopback)	Execution Window:	09:00 – 17:00 (1 Single Day)
Project Client:	Ingredion (Thailand) Co., Ltd.	Venue / Location:	xDev Workshop Integration Center
Target Enclosure:	Main Control Panel MCP-01 (TAMCO)	Target Chassis:	Chassis C1, C2, C3, C4 (52 Slots)
Total I/O Tested:	480 DI, 256 DO, 192 AI, 8 AO (936 Total)	IS Barriers:	Pepperl+Fuchs K-System (Chassis C4)

1. C1-C4 Architecture & 1-Day Site Visit Concept

The plant control core resides within MCP-01, consisting of four 13-slot ControlLogix chassis (C1, C2, C3, C4) linked by a 100Mbps ODVA Device Level Ring (DLR) and dual Cisco Catalyst 9300 Core switches.

Chassis	Role / Subsystem	Key Hardware Components	I/O Count	Destination Field Boxes
Chassis C1	Master CPU & Core I/O	1756-L950TPSXT, 3x EN4TR, 4x DI, 2x DO, 2x AI	128 DI / 64 DO / 32 AI	Central Area (CA1), JB-401, JB-601
Chassis C2	Process Remote I/O	1756-EN4TR, 5x DI, 3x DO, 4x AI	160 DI / 96 DO / 64 AI	Feed Prep, JB-402, JB-602, JB-607
Chassis C3	Extended Process RIO	1756-EN4TR, 3x DI, 2x DO, 4x AI, 1x AO (OF8)	96 DI / 64 DO / 64 AI / 8 AO	Filter Bags, JB-606, 612, 618
Chassis C4	Hazardous Area / IS RIO	1756-EN4TR, 3x DI, 1x DO, 2x AI + P&F Barriers	96 DI / 32 DO / 32 AI (All IS)	IS-JB-603, 608, 612, 618

2. Execution Principles for 1-Day Completion

- 100% Pre-FAT Internal Readiness:** Cold loop check, point-to-point continuity, and loopback simulation are fully completed prior to 09:00.
- Representative Audited Verification:** Client witnesses core failover, safety trips, and representative loop samples across C1-C4.
- Automated Closed-Loop Cross-Wiring:** Direct dry contact loopback from DO relays to DI terminals enables 100% wiring verification with automated auto-generated test logs (Detailed on Page 3).

3. Master FAT Timetable (Focus on C1–C4: 09:00 – 17:00)

Time	Dura- tion	Milestone / Agenda Item	Key Activities (Focus on Chassis C1–C4)	Responsible
09:00 - 09:15	15 min	Client Arrival & Welcome	Client reception, visitor badge, safety induction, ESD protection check before entering MCP-01 test bay.	Safety Lead / Host
09:15 - 09:30	15 min	Detail on FAT (Kick-off)	15-minute FAT Scope Briefing: C1–C4 DLR ring architecture, testing boundary, acceptance rules, Cat A vs B punch categorization.	Project Manager / Lead Automation
09:30 - 10:45	75 min	Station 1: C1 & C2 Core Audit	- C1 Master CPU (Slot 0: 1756-L950TPSXT) status, Cisco 9300 HSRP failover, Stratix DLR cable break test (recovery < 3ms). - C1 & C2 DI loop toggle simulation (1756-IB32) via TB-24VDC rail; DO relay forcing (1756-OB32 -> TBRL -> TB-0VDC).	Lead PLC Eng / Client Inspector
10:45 - 12:00	75 min	Station 2: C1 & C2 Analog Loops	- C1 (C1S10-S11) & C2 (C2S9-S12) 4-20mA calibration audit (Fluke 789 5-point test: 4, 8, 12, 16, 20mA). - Differential input circuit audit (IN-x & i RTN-x) on 1756-IF16 cards; zero crosstalk & shield grounding verification.	Instrumentation Lead / Client Rep
12:00 - 13:00	60 min	Lunch Break & Discussion	Executive hot luncheon served in private dining room. Mid-day review of morning C1/C2 test logs and workflow alignment.	All Participants
13:00 - 14:15	75 min	Station 3: C3 (AO) & C4 (P&F IS)	- Chassis C3 Modulating AO (1756-OF8: 4-20mA) valve simulation; measure current with precision DMM (+/-0.5% FS). - Chassis C4 dedicated Hazardous Area RIO with Pepperl+Fuchs K-System Zener barrier audit & Line Fault wire-break alarm.	Instrumentation & IS Lead / Client
14:15 - 15:00	45 min	Station 4: SCADA & Safety Trips	- FactoryTalk View SE SCADA integration for C1-C4 tags, faceplates, alarms, and historical trends. - Critical safety interlocks: Chamber high-temp trip, burner gas lockout, and Master E-Stop de-energization (< 50ms).	SCADA Lead / Process Eng
15:00 - 15:30	30 min	Coffee Break & Consolidation	Afternoon coffee break, pastries and refreshments. Engineering team consolidates C1-C4 test sheets into preliminary Punch List.	Hospitality / Eng Team
15:30 - 16:30	60 min	Summary Meeting & Comments	Presentation of C1-C4 test results & 100% pass rate. Detailed review of Client Comments. Categorize Punch List (Cat A vs B).	PM, Leads & Client Delegation
16:30 - 17:00	30 min	FAT Sign-off & Departure	Formal signing of FAT Certificate of Acceptance for MCP-01 (C1-C4). Review packing protection, dispatch plan & departure.	Signatories & Directors

4. Punch List Protocol & Acceptance Criteria (C1–C4 Focus)

- Category A (Critical / Blocker):** Safety hazard, hardware damage risk, or core architecture failure. Must be rectified and re-verified prior to dispatch.
- Category B (Minor / Site Item):** Cosmetic touch-up, drawing label revision, or minor spare part addition. Recorded with assigned owner and due date; does not impede shipping.
- FAT Acceptance Rule:** System is accepted when 100% of tested loops pass, network redundancy operates seamlessly, and zero Category A items remain open.

FACTORY ACCEPTANCE TEST (FAT) ENDORSEMENT – MCP-01 (CHASSIS C1, C2, C3, C4)

Project: Ingredient Kalasin — SPRINT 18K TPA Spray Dryer Plant | Ref: KAL-FAT-AGN-2608-C1C4

☐ C1–C4 Accepted without Reservation

☐ C1–C4 Accepted with Category B Punch Items

☐ Rejected

Representative of Client / Owner
Ingredient (Thailand) Co., Ltd.
Date: _____

Representative of Kalasin Engineering
Lead Automation / System Integrator
Date: _____

SPECIAL TESTING METHODOLOGY: AUTOMATED DI / DO LOOPBACK SIMULATION

Temporary Dry Contact Cross-Wiring from Interposing Relay DO to DI Input Terminal

100% Comprehensive Wiring & Hardware Validation with Autogenerated Verification Report

1. Testing Objective & Closed-Loop Principle

To verify with 100% certainty that the wiring from the PLC DO Module to the Relay Coil (A1/A2), the Relay Dry Contact (11/14), the Relay Return Line (TB-0VDC), and the DI Field Terminal to the PLC DI Module are completely sound, an Automated Closed-Loop Cross-Wiring Routine is deployed:

CLOSED-LOOP CIRCUIT ARCHITECTURE:

[PLC 1756-OB32 Output] → [Wire to Relay Coil A1] → [Relay Coil A2 to TB-0VDC]
↓ (Relay Coil Energizes & Closes Mechanical NO Contact 11 / 14)
[TB-24VDC Rail] → [Relay Contact 11/14] → [Temporary Loopback Jumper] → [DI Terminal TBDI] → [PLC 1756-IB32 Input]

2. Automated Step-by-Step Execution Sequence

- Automated Trigger:** The Python Automation Test Engine / Studio 5000 FAT Routine commands the target DO channel to Force ON (D0 = 1).
- Hardware Response:** The 1756-OB32 pin fires 24VDC to Relay Coil terminal A1. The interposing relay coil energizes, pulling down the mechanical armature and closing the Normally Open (NO) dry contact terminals 11 and 14.
- Loopback Feedback:** Positive +24VDC from the TB-24VDC distribution rail passes through closed contact 11/14, traverses the temporary cross-wiring jumper, and enters the paired DI terminal block (TBDI).
- DI Sensing & Latency Check:** The corresponding 1756-IB32 digital input channel detects +24VDC and transitions from 0 to 1. The test engine senses this feedback transition, validates that response latency is within limit ($\Delta t < 200$ ms), and marks the loop as **PASS**.
- De-energization Check:** The test engine forces D0 = 0. The relay de-energizes, the contact opens, and the DI bit drops back to 0 confirming no welded contacts or sticky relays.
- Auto-Report Generation:** The results (DO Tag, Relay Tag, DI Tag, Timestamp, Response Latency, and Verdict) are automatically logged into the master database and rendered into an official PDF/Excel report.

3. Core Technical Value & Proof of Quality

- 100% Wiring Verification:** Validates all 4 segments simultaneously: (1) PLC DO to Relay, (2) Relay Coil 0V return, (3) Relay contact switching, and (4) DI Terminal to PLC DI module.
- Zero Human Forcing Error:** Completely eliminates manual wire-tapping or human misreading of terminal numbers. Tests up to 256 DO channels in under 5 minutes.
- Deterministic Traceability:** Generates an autogenerated, timestamped test certificate signed off by the lead automation engineer.

4. Sample Autogenerated Loopback Test Log (Chassis C1-C4 Excerpt)

Loop	DO Slot / Pin	Relay Tag (A1/A2)	Paired DI Terminal	DI Slot / Pin	Latency	State	Verdict
LP-01	C1S8 - Pin 1	P1-TBRL1-1 (A1/A2)	P1-TBDI1-1 (11/14)	C1S4 - Pin 1	28 ms	1 -> 1	PASS
LP-02	C1S8 - Pin 2	P1-TBRL1-2 (A1/A2)	P1-TBDI1-2 (11/14)	C1S4 - Pin 2	31 ms	1 -> 1	PASS
LP-03	C1S8 - Pin 3	P1-TBRL1-3 (A1/A2)	P1-TBDI1-3 (11/14)	C1S4 - Pin 3	26 ms	1 -> 1	PASS
LP-04	C2S6 - Pin 1	P2-TBRL1-1 (A1/A2)	P2-TBDI1-1 (11/14)	C2S1 - Pin 1	34 ms	1 -> 1	PASS
LP-05	C2S6 - Pin 2	P2-TBRL1-2 (A1/A2)	P2-TBDI1-2 (11/14)	C2S1 - Pin 2	29 ms	1 -> 1	PASS
LP-06	C3S4 - Pin 1	P3-TBRL1-1 (A1/A2)	P3-TBDI1-1 (11/14)	C3S1 - Pin 1	33 ms	1 -> 1	PASS
LP-07	C4S4 - Pin 1	P4-TBRL1-1 (A1/A2)	P4-TBDI1-1 (11/14)	C4S1 - Pin 1	37 ms	1 -> 1	PASS

Generated by Kalasin Engineering Automated Loopback Test Engine v2.4 | Master Report: KAL-FAT-TR-DIDO-2608